

Saurabh Verma

Quality Assurance Engineer

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SUMMARY

Results-oriented Mechanical Engineer with 7+ years of cross-functional experience in mechanical design, quality assurance, and industrial project execution within manufacturing and automation domains. Demonstrated success in DFMEA, GD&T, Lean Six Sigma, tolerance analysis, reliability testing, and supplier quality management. Adept at using SolidWorks, ANSYS, AutoCAD, Primavera P6, and Power BI to drive design optimization, product reliability, and process improvement. Deep understanding of ASME, ISO 9001, OSHA, RoHS, and REACH standards ensuring compliance, safety, and cost efficiency across the product life cycle.

PROFESSIONAL EXPERIENCE

Mechanical Engineer (Quality Assurance) | Honeywell | USA

May 2025 – Current

- Spearhead mechanical design validation and QA initiatives for HVAC and industrial control assemblies, ensuring adherence to ASME Y14.5, ISO 9001, and OSHA standards across production lines.
- Conduct Root-Cause Analysis (RCA), 8D Reports, and DFMEA reviews using Minitab and SolidWorks to eliminate recurring failures, achieving 27 % reduction in rework and saving \$450 K annually.
- Drive Design for Manufacturability (DFMA) improvements by collaborating with design and manufacturing teams, cutting production cycle time by 22 % and enhancing assembly precision.
- Lead supplier quality audits, PPAP documentation, and APQP activities, increasing vendor quality performance by 15 % and achieving consistent PPM targets (< 350 PPM).
- Develop and automate QA dashboards (Power BI / Excel VBA) for tracking OEE, FPY, and SPC metrics, enabling data-driven corrective actions and real-time defect monitoring.
- Execute tolerance stack-up, CMM inspection, and MSA studies to validate mechanical tolerances ± 0.01 mm, ensuring repeatability and measurement accuracy > 95 %.
- Implement Lean Six Sigma projects (DMAIC) process standardization and waste elimination, reducing QA inspection lead time by 30 %.
- Oversee thermal, vibration, and endurance testing of mechanical components, utilizing ANSYS to ensure design safety factors > 2.0 under operational stress.
- Collaborate with R&D, procurement, and production teams for NPI validation, DFMEA reviews, and supplier qualification, maintaining full compliance with global Honeywell standards.
- Mentor juniors in QA methodologies, DFMA principles, and CAD interpretation, boosting team productivity and technical capability.

Assistant Manager (Projects) | Bansal Steel and Power Ltd | India

March 2019 – Aug 2023

- Directed mechanical installation projects for conveyors, furnaces, and crane assemblies worth \$5 M+, coordinating layout, fabrication, and commissioning activities.
- Prepared and reviewed equipment layout drawings, isometric piping designs, and 3D models using AutoCAD and SolidWorks to ensure dimensional accuracy and interference-free fit.
- Implemented Lean manufacturing and Six Sigma projects to reduce material waste and improve process flow, achieving 25 % throughput increase.
- Conducted supplier and contractor performance audits, establishing PPAP and FMEA systems that cut delivery delays by 20 %.
- Led FMEA and preventive maintenance planning for critical equipment, raising plant uptime to 98 % and reducing unplanned downtime events by 30 %.
- Managed EHS compliance per OSHA and ISO 14001 standards during installation and shutdown activities, ensuring zero recordable incidents.
- Oversaw procurement and inventory control for mechanical components using SAP PP/QM, reducing material stock variance by 12 %.
- Collaborated with the design team on process optimization and equipment upgradation to improve energy efficiency and reduce O&M costs by 10 %.
- Applied Earned Value Management (EVM) to track schedule adherence and budget performance, maintaining CPI/SPI indices ≥ 1.0 .
- Reported progress and technical challenges to project stakeholders, facilitating timely decision-making and alignment with organizational objectives.

Project Planning Engineer | Preet Machines Ltd | India

Aug 2018 – March 2019

- Managed end-to-end planning and execution of steel rolling mill projects valued at \$10 M+, ensuring cost, schedule, and scope alignment through Primavera P6 and MS Project.
- Created detailed project schedules, WBS, and EVM dashboards, achieving on-time delivery performance > 96 % across 6 major projects.
- Coordinated with design and manufacturing departments to prepare AutoCAD and SolidWorks drawings, BOMs, and material specifications for fabrication.
- Implemented Lean principles and 5S audits within assembly areas, reducing non-value-added activities by 20 % and enhancing floor productivity.
- Conducted thermal stress and static FEA simulations in ANSYS to validate mechanical integrity, ensuring stress levels remained > 1.5 safety factor under load.
- Established QA inspection checkpoints and developed QC plans for vendor and in-house fabrication, lowering installation defects by 18 %.
- Performed capacity planning, resource allocation, and procurement coordination, optimizing material lead time by 15 % and ensuring

- supplier OTD > 98 %.
- Prepared progress and variance reports, using Power BI dashboards to visualize CPI / SPI performance and critical path delays.
- Supervised site installation and commissioning of mechanical assemblies, including hydraulic and pneumatic systems, ensuring safety compliance under OSHA guidelines.
- Collaborated with teams to standardize engineering change management (ECM) processes, reducing design revisions by 25 %.

Assistant Engineer (Mechanical) | Pushp Rings Pvt Ltd | India.

Aug 2016 – Aug 2018

- Assisted in production planning and process optimization for automotive ring components and forged metal parts, supporting a high-volume manufacturing environment.
- Operated and maintained CNC machines and conventional lathes, reducing tool change time by 15 % through SMED principles.
- Created 2D/3D drawings in AutoCAD and SolidWorks for fixture design and component modifications, enhancing fit accuracy by 10 %.
- Performed in-process and final inspection using CMM, micrometers, and calipers, ensuring GD&T tolerances within ± 0.01 mm.
- Supported QA team in conducting SPC and MSA analysis to monitor process capability ($Cpk > 1.67$).
- Participated in root-cause analysis and corrective action implementation for recurring defects, lowering rejection rate by 20 %.
- Implemented 5S and Kaizen initiatives on the shop floor, boosting workspace efficiency and operator safety.
- Collaborated with tooling and maintenance teams to improve machine utilization from 78 % to 90 %.
- Supported new product trial runs and validation testing per customer specifications and ISO 9001 requirements.
- Documented process flow charts, control plans, and checklists for production standardization and training purposes.

TECHNICAL SKILLS

- Design & Analysis:** SolidWorks, AutoCAD, CATIA V5, ANSYS, Creo, MATLAB, Simulink
- Quality Tools:** FMEA, PPAP, 8D Reports, MSA, SPC, RCA, CAPA, DFMA, PFMEA
- Manufacturing Processes:** CNC Machining, Casting, Forging, Welding, Assembly Line Optimization, 3D Printing
- Project Management:** MS Project, Primavera P6, Agile (Scrum), Lean Six Sigma (Green Belt), Kanban, 5S, Kaizen
- Testing & Validation:** Vibration Testing, Thermal Analysis, Fatigue & Failure Testing, CMM, Tolerance Stack-up
- Software & Data Tools:** Excel VBA, SQL, Tableau, Power BI (for QA KPIs and OEE Dashboards)
- Standards & Regulations:** ASME Y14.5, ISO 9001, ISO 14001, OSHA Safety, RoHS, REACH
- Soft Skills:** Cross-Functional Collaboration, Problem Solving, Analytical Thinking, Documentation, Continuous Improvement

EDUCATION

- Masters in Mechanical Engineering** | University of Colorado Denver, Colorado, USA
- Aug 2023 – May 2025**
- Bachelors in Mechanical Engineering** | RTU KOTA, India
- Aug 2012 - May 2016**